

Question Paper

Exam Date & Time: 22-Nov-2025 (02:00 PM - 05:00 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU
FIRST SEMESTER M.TECH. EXAMINATIONS - NOVEMBER 2025

Digital CMOS IC Design [ECE_5151]

Marks: 50

Duration: 180 mins.

PART-A

Answer all the questions.

- 1) For a Boolean logic given: $Y = (A + B.C.A')$ which has a drive strength of x16 at the output stage, (5)
A) Draw the CMOS transistor level circuit and size the transistors using logical effort method and stage ratios.

Note:

- Do not apply any Boolean deduction methods to reduce the equation. Do it as is.
- Consider $W_{pmos}=2W$ and $W_{nmos}=W$ to get equal pull-up and pull-down current in an inverter while driving the same sized inverter.
- Use Stage ratio as 1:4
- A' should be generated as part of the circuit and transistors to be sized appropriately.

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- B) For a Boolean logic given: $Y = (A + B.C')'$ in which the C' is internally generated using signal C. (5)
Draw the CMOS transistor level circuit and respective Stick Diagram which is optimized to keep minimum parasitic capacitances. Show the Euler's path used to arrive at the optimized drain/source sharing and ensure to have least internal drain capacitance at node Y.

Note:

- Do not apply any Boolean deduction methods to reduce the equation. Do it as is.
- Use colour code as below for stick diagram

- a. Power - VDD and VSS - Red
- b. Poly - Green
- c. Metal - Dark Blue
- d. Oxide - Use Rectangle with Orange highlight
- e. Contacts - Use 'X' in black

Course Outcome: "CO1" / Cognitive Level: "Analysis"

- 2) Explain CMOS Twin Well Fabrication method using a Step by Step Approach to achieve a cross section of CMOS Inverter. Explain the reason for N^+ implant in Nwell and P^+ implant in P-Well. (4)

A)

Course Outcome: "CO2" / Cognitive Level: "Understand"

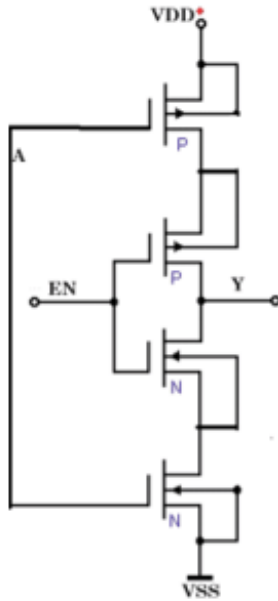
B)

Explain Photo Lithography and different types of Photo-resists.

(2)

Course Outcome: "CO2" / Cognitive Level: "Understand"

- C) Draw the Cross section with proper terminal representations and connections of below circuit using Deep N-Well Process for CMOS Fabrication. (4)



Course Outcome: "CO2" / Cognitive Level: "Application"

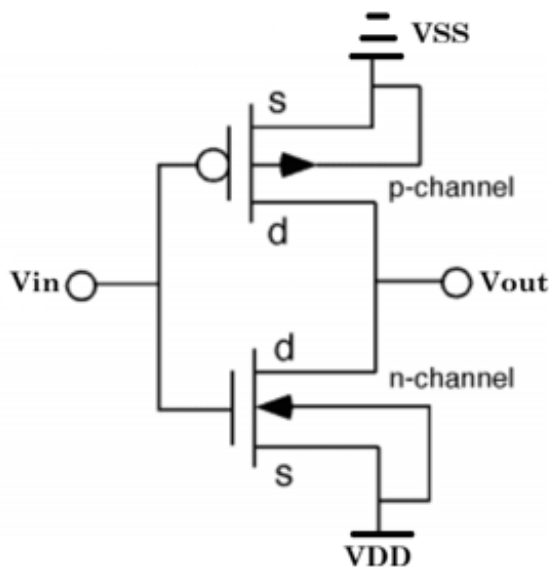
- 3) Explain NMOS Operation with appropriate cross section images and respective equations for different regions of operation and graphs of (I_{ds} Vs V_{ds} with different V_{gs}) and (I_{ds} Vs V_{gs} for constant V_{ds}) (4)
- A)

Course Outcome: "CO3" / Cognitive Level: "Understand"

- B) Explain Voltage Transfer Characteristics of a CMOS Inverter with different regions of operation. (4)
Consider the switching point of the ideal inverter is at $V_{DD}/2$, if we increase the width of only PMOS by keeping NMOS width constant, what happens to the switching point. Show it graphically.

Course Outcome: "CO3" / Cognitive Level: "Application"

- C) Write the Truth Table and Explain the disadvantages of below circuit. (2)



Course Outcome: "CO3" / Cognitive Level: "Understand"

- 4) Write the Truth Table of a 2:1 MUX and Explain the working of it using both restoring logic (Tri-State) and non-restoring logic (TG-based). (4)

A)

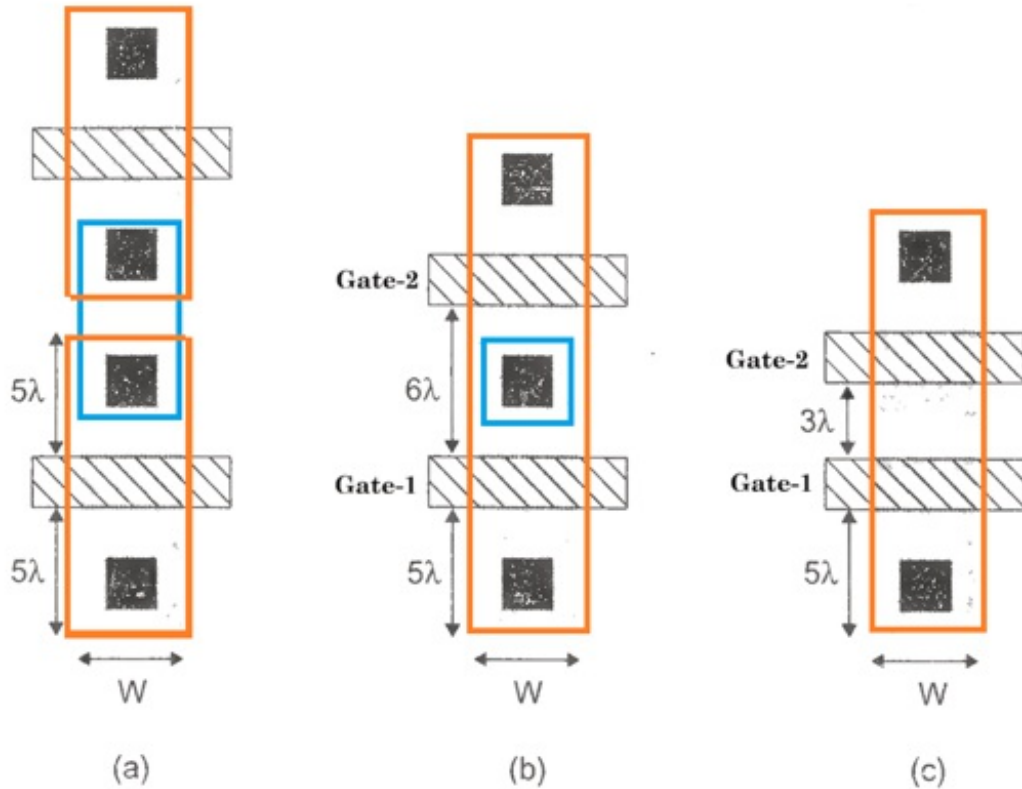
Course Outcome: "CO4" / Cognitive Level: "Understand"

- B) Explain the Advantages and Disadvantages of different layout styles shown below. In Which Specific Scenario should we use the layout style as in Figure (a) (4)

Figure (a) is Separated Diffusion connected by Metal and contacts.

Figure (b) is Shared Contacted Diffusion

Figure (c) is Non-Contacted shared diffusion.



Course Outcome: "CO4" / Cognitive Level: "Analysis"

- C) Explain the Advantages and Disadvantages of different layout styles shown below. What will be the effective Width and Length of the below circuits. (2)

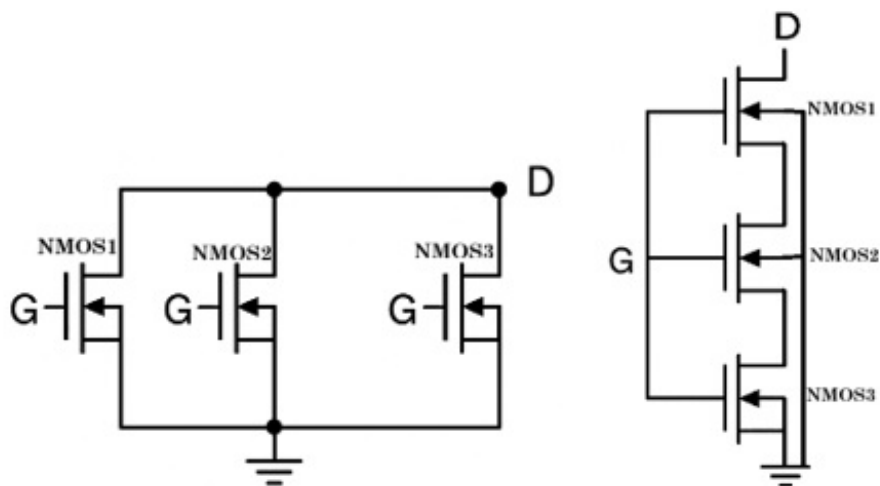
NMOS1 - $W=2\mu\text{m}$, $L=100\text{nm}$

NMOS2 - $W=4\mu\text{m}$, $L=100\text{nm}$

NMOS3 - $W=6\mu\text{m}$, $L=100\text{nm}$

$W_{\text{parallel}} = ?$ $L_{\text{parallel}} = ?$

$W_{\text{series}} = ?$ $L_{\text{series}} = ?$



Course Outcome: "CO4" / Cognitive Level: "Understand"

- 5) Explain the phenomenon of Electro-migration. What layout and routing techniques can reduce electro-migration risk in VLSI design? (4)

A)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- B) Explain Latch-up in detail using a cross-section diagram of a CMOS inverter. Show the parasitic devices that are formed in the cross-section. (4)

Course Outcome: "CO5" / Cognitive Level: "Understand"

- C) Explain the working of a 6-T memory cell. (2)

Course Outcome: "CO5" / Cognitive Level: "Understand"

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Event Name	Subject
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058-END SEMSTER-ODD-2025-PG ECE_5151:Digital CMOS IC Design

#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* **NA** - CO Not Selected

Event Name	Subject
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058-END SEMSTER-ODD-2025-PG ECE_5151:Digital CMOS IC Design

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	9	28(56%)
3	Application	2	8(16%)
4	Analysis	3	14(28%)
5	Create	0	0(0%)

6	Evaluate	0	0(0%)
	Total	14	50 (100%)